

Day - 1 Intro

Topic 1: What is a Database?

A database is an organized collection of data.

Example:

Student_ID	Name	Branch
1	Rutvik	CSE
2	Kavya	AIML

This table is stored inside a database.

Topic 2: What is RDBMS?

RDBMS = Relational Database Management System

Examples:

- PostgreSQL
- MySQL
- Oracle
- SQL Server

Data is stored in tables and relationships are maintained using keys.

Example:

Students Table

Student_ID	Name
1	Rutvik

Courses Table

Course_ID	Course
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101 SQL

Enrollment Table

Student_ID	Course_ID
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1	101
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This relationship is the heart of RDBMS.

Topic 3: SQL vs NoSQL

SQL	NoSQL
Tables	Documents
Fixed Schema	Flexible Schema
ACID	BASE
PostgreSQL	MongoDB
Joins	Embedded Data

Use SQL when:

- Banking
- Healthcare
- ERP
- LMS
- Analytics

Use NoSQL when:

- Social Media
 - Chat Apps
 - Large-scale logs
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Topic 4: SQL Syntax

```
SELECT * FROM students;
```

Meaning:

SELECT → choose data

- → all columns

FROM → source table

students → table name

Topic 5: Data Types

Numeric

INT

BIGINT

DECIMAL

FLOAT

Text

CHAR

VARCHAR

TEXT

Date

DATE

TIME

TIMESTAMP

Boolean

BOOLEAN

Topic 6: First Query

Suppose table:

students

id	name	age
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1	Rutvik	25
---	--------	----

2	Kavya	24
---	-------	----

Query:

```
SELECT * FROM students;
```

Output:

1 Rutvik 25

2 Kavya 24

Practice Set 1

Answer these without looking at notes.

Q1

Which command retrieves data?

A. INSERT

B. DELETE

C. SELECT

D. UPDATE

Q2

Which database is relational?

A. MongoDB

B. PostgreSQL

C. Redis

D. Cassandra

Q3

Which data type stores text?

- A. INT
 - B. BOOLEAN
 - C. VARCHAR
 - D. DATE
-

Q4

What does * mean in:

SELECT * FROM students;

Q5

Write a query to display all data from a table called **employees**.

?

To-Do work

1. Install PostgreSQL.
2. Install pgAdmin.
3. Create a database called:

sql_mastery

4. Create a table:

students

with columns:

id
name
age
city

5. Insert 10 records.

